

23-13 How does decarbonization change the fiscal equation?

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ABSTRACT

This paper explores the fiscal implications for countries of global climate mitigation in the medium term. If climate action is unilateral, it might be limited in scope and rely more on subsidies and spending to avert political constraints. This can put fiscal sustainability at risk. Coordinated carbon pricing or other mitigation policy can more effectively put the world on a path to 1.5 to 2°C above pre-industrial temperatures, as agreed in Paris in 2015, while helping manage fiscal and political constraints. Coordination could be initiated by large players, such as China, the US, India, the African Union and the European Union. We find that the implications for fiscal revenues over time are shaped by a combination of rising carbon prices, the gradual erosion of existing fuel tax bases and possible revenue sharing arrangements. Public spending rises during the transition to build green public infrastructure, promote innovation, and support clean technology deployment, although much of this spending could be more efficiently financed through higher sectoral prices and taxes rather than through the general budget. Countries will also need funds for compensating vulnerable households, industries, and poor countries. With well-designed climate-fiscal policy relying on carbon pricing, global decarbonization will have anything from moderately positive to moderately negative impacts on fiscal balances in high-income countries. For middle and low-income countries, net fiscal impacts are generally positive and significant. Hence, as mitigation strategies improve fiscal balances, they can accommodate development spending needs. Revenue sharing at the global level would make an historical contribution to breaching the financial divide between rich and poor countries.

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I. INTRODUCTION

The planet is warming. The summer of 2023 was the hottest year on record. According to the World Meteorological Organization, temperatures are likely to increase by more than 1.5°C above pre-industrial levels already within the coming five years.¹ If unmitigated, climate change is expected to increase temperatures relative to pre-industrial levels by more than 3°C by 2100.² This would have devastating effects on macroeconomic stability through lower productivity in agriculture, fishing, and work in non-climatized locations; more frequent disruption of activity and destruction of productive capital due to extreme weather events, natural disasters, and rising sea levels; diversion of resources toward adaptation and reconstruction; increased morbidity and mortality due to more prevalent infectious diseases and natural disasters; and greater risks of migration and human conflict. Given the strong interconnectedness of the global economy, no country is likely to remain unscathed, even if the worst warming impacts are initially concentrated in hotter regions.

In 2015 in Paris, countries agreed to reduce greenhouse gases (GHG) emissions to “holding the increase in the global average temperature to well below 2°C above pre-industrial levels” and ideally to 1.5°C (Black et al. 2023c; Hassler et al. 2023; Gourinchas et al. 2023). To achieve this, they have defined Nationally Determined Contributions (NDCs) for carbon,³ commonly defined as emission reduction targets in or around 2030 relative to some benchmark. More recently, several countries have also committed to targets for zero net carbon emissions by around midcentury (i.e., a zero balance of gross carbon emissions and negative emissions from carbon sinks and through technology, e.g., carbon capture and storage). Unfortunately, while several countries have achieved progress in reducing carbon emissions, there remain large ambition- and implementation gaps for 2030 against the agreed temperature goals (Black et al. 2023c; Gourinchas et al. 2023). Section II of this paper provides further details on current mitigation targets, policies, and emission impacts of these policies.

Greater policy ambition and effort is therefore needed to boost investment in low-carbon technologies, spur innovation and change the behavior of households and firms. Estimates suggest that, to get on track to net zero emissions by midcentury, the total necessary investment runs into several trillions of US\$. An important question is: Who will pay for this? Will private households and firms bear the brunt of these costs, incentivized by government regulations and policies? Or will the public sector pay for a significant share of it and shift the burden to future generations by raising public debt? Section III of this paper argues that governments face a policy trilemma in choosing its policy mix, reflecting a three-way trade-off between (i) meeting climate policy objectives; (ii) keeping public debt sustainable; and (iii) managing political constraints. More specifically, relying more on public spending policies (through subsidies and green public investments) might be politically easier to meet a country’s climate goals but will more likely be inefficient and fiscally unsustainable. On the other hand, carbon pricing is more efficient and has the

¹ <https://public.wmo.int/en/media/press-release/global-temperatures-set-reach-new-records-next-five-years>

² Central estimate per the IPCC’s scenario ‘SSP2-4.5’ though the range is large at 2.1-3.5°C (IPCC 2022).

³ The focus here is on reducing carbon emissions, which is responsible for 75 percent of total GHGs. For a discussion of methane emissions (17 percent) and policies to address them, see Parry et al. (2022).

potential to improve fiscal outcomes but can meet more political resistance. Scaling back climate ambition could relax both fiscal and political constraints but violates climate commitments. Hence, governments will need to navigate climate neutrality with debt sustainability and political feasibility.

Forceful and urgent global action could ease this policy trilemma and help achieve the world's climate goals agreed in Paris. Unilateral climate action should still be encouraged to keep up momentum towards coordinated global action and is found to have important positive spillover effects (Linsenmeier, Mohommad, and Schwerhoff 2022). However, it can be difficult for countries to adequately scale up mitigation policies unilaterally in light of the global nature of the problem. For instance, each individual country will be concerned about the lack of ambition in other countries and possible effects of unilateral action on its competitiveness. In the absence of global coordination, unilateral action will therefore likely be more limited and characterized by insufficient mitigation ambition and policy implementation, greater emphasis on regulatory and subsidy approaches, and limited international support to finance clean energy transitions in developing countries. This is a critical aspect as high-income countries cannot adequately mitigate global climate change by acting alone, given the large share of emissions and projected growth in middle- and low-income countries. For instance, by 2030 the EU and the US together will emit a little over 20 percent of global emissions, while China and India will be responsible for almost 40 percent (Black et al 2023c). Achieving the world's climate goals would therefore be facilitated by a global side agreement on climate policy that complements the Paris Agreement and overcomes the obstacles to unilateral action.

This paper explores the fiscal equation under a concrete and equitable global side agreement scenario, chosen to close global mitigation gaps in 2030 and be consistent with a path of 2°C warming (the upper end of the Paris temperature goals) (Section IV). We consider a combination of policies and assess their fiscal implications. On the revenue side, we look at a mix of carbon pricing and non-pricing policies and consider the fiscal revenue effects from the erosion of existing fuel tax bases as the economy decarbonizes (Section V). On the spending side, we discuss public investment in green infrastructure, subsidies for low-carbon technological innovation and deployment (Section VI), and support for vulnerable households and affected firms, industries, and locations (Section VII). Subsequently, we discuss the implications of possible international revenue sharing arrangements (Section VIII). Section IX puts the pieces of the fiscal equation together. Finally, Section X concludes.

II. CURRENT AND PLANNED MITIGATION POLICIES

We start by taking stock of existing and planned mitigation policies at economywide and sectoral levels for G20 countries, as specified in their Nationally Determined Contributions (NDC) and other government documents. Policies are summarized in more detail in Black et al. (2023c). This forms the baseline for subsequent sections. Overall, our estimates put the combined reduction in emissions as a result of these policies, relative to a no-policy counterfactual for 2030, at 11 percent across the G20. Such a reduction falls significantly short of the 25–50 percent reduction by 2030 needed to achieve the Paris Agreement's temperature goals of, respectively, 2°C and 1.5°C warming.

Carbon pricing schemes (at various levels of government) are operating in twelve G20 countries.⁴ In eight cases, these cover more than 50 percent of national GHGs (Canada, France, Japan, Korea, Mexico, South Africa, Germany, and the UK). For example, emissions trading systems (ETSs) at the European Union (EU) level and in the United Kingdom (UK) apply to emissions from power generation and industry. Japan and South Africa apply carbon taxes midstream on fuel supply, while Korea's ETS covers emissions for power and industry. Canada requires provinces and territories to implement a minimum carbon price through a carbon tax or ETS.

For power generation, nearly all G20 countries have renewable energy targets and eight have targets for scaling back coal generation (including in five cases a complete ban on or before 2030). Common policies for the sector are feed-in tariffs, renewable portfolio standards, and investment or production tax credits. The analysis here assumes sectoral targets are met—it would be difficult, analytically, to disentangle the impacts of individual, overlapping measures.

For transportation, aside from fuel taxes, CO₂ emission rate or fuel economy standards apply nationally in nine G20 countries and at the EU level. Fifteen countries have targets for phasing in electric vehicles (EVs) or phasing out internal combustion engine vehicles (ICEVs). Feebates involving fees for the purchase of emissions-intensive vehicles and subsidies for relatively clean vehicles, apply in some form in nine countries. Emission rate standards, feebates, and EV sales requirements aid the transition to clean vehicle fleets, though they do not promote faster retirement of existing, high-emission vehicles nor reduce vehicle kilometers (km) travelled—again, disentangling their individual CO₂ reductions would be difficult analytically.

For buildings, France, Germany, Italy, and Japan have targets for reducing energy use from the total building stock (by 25-44 percent between 2020 and 2030), while nine other G20 countries have targets for new buildings to produce approximately zero emissions by 2030 or later. Countries typically use multiple instruments such as design codes to reduce energy consumption, incentives for insulation and other retrofitting, certification programs for meeting green criteria, clean fuel requirements (e.g., phasing out fossil fuel heating in new buildings), and efficiency standards for appliances. Combinations of these policies reduce emissions, but less effectively to the extent they apply only to new as opposed to all capital, especially given the gradual turnover of the building stock.

Industrial sectors are generally subject to lighter sectoral emissions targets/policies than other energy-intensive sectors. Only five G20 countries have binding emissions targets for industry. Regulations like tradable performance standards (TPSs)⁵ and fiscal incentives like clean technology subsidies are less common for this sector. An exception is Canada where TPSs apply to industrial emissions in many provinces and territories, while China intends to expand its (more restricted form of) TPS for power generation to industry.

⁴ Globally, there are 73 national, subnational, and regional carbon pricing schemes in place, covering almost one quarter of global emissions. At the same time, the average price in several of these schemes is low at on average around \$20 per ton of carbon.

⁵ Under TPSs, firms are required to meet a standard for average carbon emissions per kilowatt-hour across power generation plants or per ton of steel. Those that fall short of the standard can purchase credits from firms that exceed the standard. These standards require fluid markets for trading credits and are less practical for some sectors, such as forestry and residential buildings.

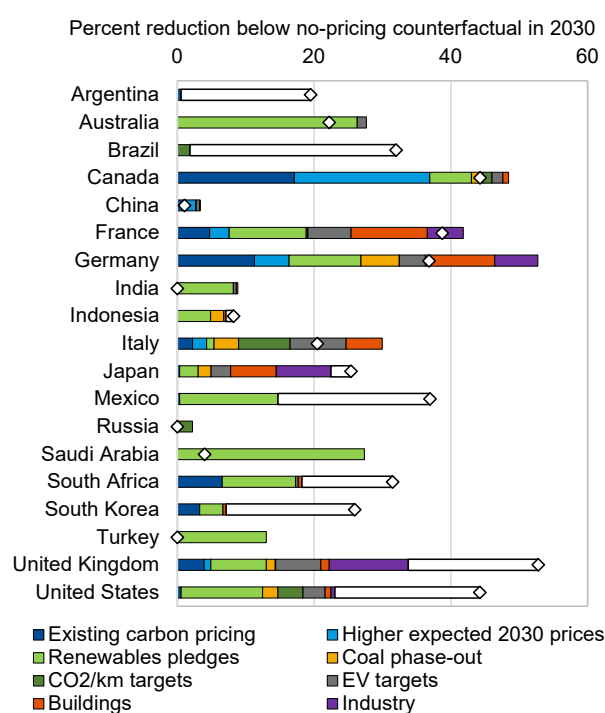
The combined effect from the above policies and targets up to 2022 varies substantially across countries (Figure 1). CO₂ reductions relative to a 2030 baseline with no carbon policy range between 20 and 50 percent in 11 countries and are less than 20 percent in eight cases. Additionally, countries vary significantly in their choice of instrument and relative contribution of sectoral targets. Renewables targets make a significant contribution to reductions in the policy mix in 15 cases and explicit carbon pricing contributes substantively in eight cases. The attribution of emissions reductions to individual policies and targets is ambiguous where they overlap however (e.g., for carbon pricing of power emissions and renewable generation targets) hence the total CO₂ reductions should be considered more accurate than the relative contribution of specific policies and targets.

Countries vary in the extent to which policies achieve economy-wide mitigation pledges in NDCs. Targets are not binding, or barely binding, in four cases—including two of the three largest global emitters (China and India). Some countries over-achieve NDCs with sectoral policies (Australia, Canada, China, France, Germany, Italy, Saudi Arabia)—economywide targets could be enhanced in these cases by bringing them in line with existing sectoral targets, though this relies on sectoral targets being met. In seven cases the economywide emissions reductions from policies and targets identified for this analysis fall well short of the reductions needed for NDCs.

Some big economies have recently adopted policy packages that largely rely on expenditures-based measures. These include public funding for investments in clean energy and green subsidies (or tax expenditures) to provide incentives for private investment and adoption of low-carbon technologies. For example, the Inflation Reduction Act of 2022 represents

the largest federal policy to date in the United States (US), costing nearly \$400 billion over 10 years, to tackle climate change. It envisages subsidies and tax incentives for investments in clean electricity and transmission, clean transportation and energy-efficiency upgrades in homes (see Bown and Clausing 2023; Gourinchas et al. 2023). The EU has supplemented its carbon-pricing approach by proposing a Green Deal Industrial Plan comprising tax breaks and relaxation of state aid (subsidy) rules in the coming years to boost renewable investment of the private sector. China has scaled up green public investment and subsidized the deployment of solar energy over the last decade under its Made in China 2025 initiative. Other countries, such as Canada, Japan and Korea have adopted similar policies.

Figure 1. Emissions Impacts of Current Policies and Sectoral Targets



Source: IMF staff using CPAT.

Note: The white bars indicate policies that have not been specified or could not be quantified for this exercise. Diamonds represent NDCs.

III. A POLICY TRILEMMA ON THE ROAD TO CARBON NEUTRALITY

The previous section makes clear that more policy ambition and effort is needed globally to get on track to net zero emissions by midcentury. Estimates of the total investment necessary to achieve carbon mitigation objectives (such as renewable energy, electrification, energy saving and carbon capture) vary significantly but often run into trillions of US\$. For instance, the World Energy Outlook (2022) of the International Energy Agency (IEA) reports that global energy investment, currently estimated at \$2 trillion, would more than double in a net zero scenario – an increase by more than 2 percent of global GDP.⁶

Who will pay for this? We expect the private sector to fund most of the investment expenses for decarbonization. For example, incentivized by pricing and regulatory policies, private firms will invest in renewable power generation and electrified industrial processes, while households will pay for purchases of EVs, heat pumps, and energy-efficient lighting/appliances. However, recent policy developments suggest that several governments are inclined to share in these costs by compensating households and firms for at least a portion of the investments through subsidies and tax reliefs. For instance, an increasing number of countries provide subsidies for EVs, batteries and renewable energy. And to reduce emissions from buildings, several countries require the replacement of gas boilers and other fossil-fuel heating systems by cleaner alternatives, such as heat pumps. As these are generally expensive, some governments choose to compensate households through subsidies.

The IMF's Fiscal Monitor of October 2023 explores the fiscal implications of a scenario whereby countries achieve their net zero emission objective by midcentury, thereby heavily relying on expenditure-based policy measures—thus scaling up the recent policy efforts of some large economies (IMF 2023a). Using a dynamic general equilibrium model, it simulates the effects of such a policy package on debt dynamics until 2050 for a representative advanced economy and a representative emerging market economy. In the model, the fiscal effects depend on how fiscal instruments affect growth and interest rates. For instance, expenditure-based measures will support output in the short term, but also raise deficits and add to the pressures on interest rates and government borrowing costs by raising the demand for capital and increasing the supply of government debt. Changes in the primary government balance and endogenous effects on output and interest rates determine the debt dynamics until 2050.

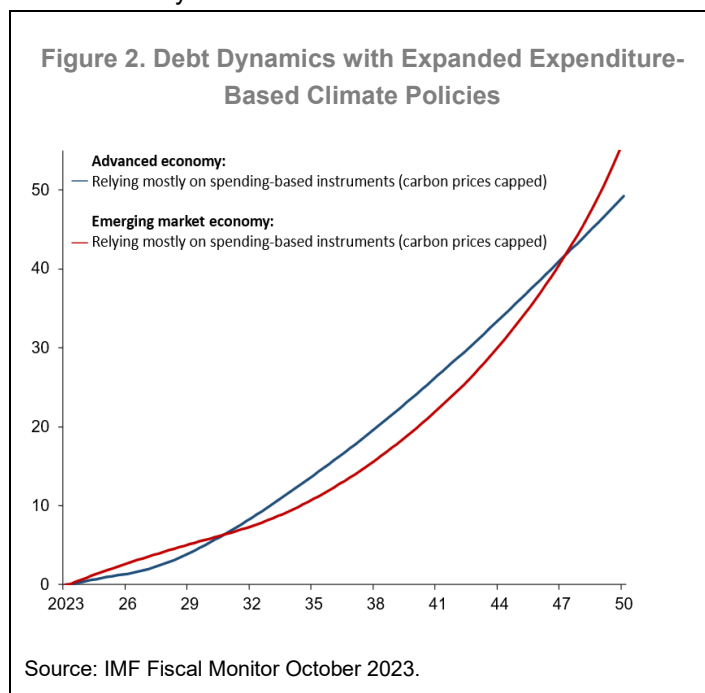
The analysis in the Fiscal Monitor operationalizes the net-zero-emissions target as an 80 percent reduction in 2023 gross emission levels by 2050 for advanced economies and by 2060 for emerging market economies, with the assumption that carbon capture and storage will offset the remaining emissions. For the representative advanced economy—which is the focus here—the simulation considers a policy package that combines a carbon price that rises to \$75 a ton by 2030 and maintains at that level until 2050. An extensive scaling up of public investments and subsidies is then employed to achieve the remaining emission reductions. If unfunded, the simulations show that this policy package leads to a rise in the debt-to-GDP ratio by 45 percentage points by 2050, in part caused by increases in government

⁶ Estimates of global investment needs vary significantly across studies, see e.g. IMF (2021). Country-specific estimates vary even more and depend on specific circumstances such as carbon intensity.

borrowing costs (Figure 2). Hence, scaling up spending (such as those recently put in place in several advanced economies) will lead to increases in debt levels that are likely unsustainable.⁷

The Fiscal Monitor therefore explores an alternative policy scenario, which combines revenue and expenditure measures to achieve net zero emissions. The policy package for the representative advanced economy includes a carbon price that increases from \$40 a ton today to \$130 by 2030 and \$235 by 2050. Revenues from carbon pricing are projected to peak around 2030 as decarbonization will gradually erode the carbon tax base. Thirty percent of the carbon revenues are used to compensate households through cash transfers. On the expenditure side, the simulations assume an increase in green public investment and green subsidies of around ½ percent of GDP, which is smaller than in the previous scenario (where spending rises by nearly 2 percent of GDP) but still a sizeable spending increase. On balance, the simulation still finds a considerably smaller increase in the debt-to-GDP ratio of around 10–15 percentage points by 2050, relative to the baseline.

Other policy scenarios could be envisaged, however, that place greater weight on efficiency and fiscal sustainability. Indeed, public spending for decarbonization could be scaled back further to cover only necessary and efficient spending on public infrastructure and technology. At the same time, greater reliance on carbon pricing and other sectoral instruments could efficiently incentivize decarbonization in the private sector. As illustrated in the Fiscal Monitor's sensitivity analysis, such a scenario would imply a reduction in the debt level by 2050.⁸ The challenge with the latter scenario might be its political feasibility, as higher energy prices are generally unpopular and can meet fierce public resistance. This points to a policy trilemma between (i) meeting the climate policy objective, (ii) maintaining debt sustainability and (iii) respecting political constraints. The high spending scenario above can be seen as a way to meet a country's climate policy goal while avoiding political resistance. However, that scenario is fiscally unsustainable. The scenario that relies most on carbon pricing



⁷ Apart from debt sustainability, this raises issues of intergenerational equity. Current generations are asked to incur a cost by investing in more expensive green technologies to curb emissions and avoid damages for the benefit of future generations. If current generations are reluctant to do so, this can be reflected in political obstacles to the green transition. Transfers from future to current generations by raising government debt could make future generations pay for climate action and relax current political opposition. Kotlikoff et al. (2021) show that a carbon tax that comes along with a well-designed package of intergenerational transfers can indeed raise welfare of current and future generations, while limiting the rise in global temperatures to 1.5°C. Yet, different views on intergenerational equity are possible. For instance, some may argue that future generations will already suffer from severe climate damages caused by past and current emissions and that imposing an additional public debt burden on them might be unfair.

⁸ For a comparison between the analysis in the Fiscal Monitor and in this paper, see the Annex.

produces positive fiscal outcomes but could cross political red lines. Scaling back policy ambition could relax both the fiscal and political constraints but violates climate commitments. Hence, governments face a three-way trade-off between achieving climate neutrality, ensuring debt sustainability, and respecting political constraints.

Studies along similar lines as in the Fiscal Monitor have been conducted for the UK and France, using country-specific assumptions and models. The Office of Budget Responsibility (2021) illustrates the fiscal implications of various scenarios for revenue and spending to achieve the net zero emissions goal in the UK in 2050. Key assumptions are made about carbon pricing, the loss of fuel excise revenue (i.e., whether its revenue loss will be offset by a new charge on road use) and the government's support for insulation and heat pumps to decarbonize residential buildings. In OBR's central scenario (without a road-user charge and with a 50 percent subsidy for decarbonization of buildings), the debt-to-GDP ratio increases by 21 percent in 2050. However, in a scenario where a road-user charge is introduced and subsidies are scaled back, the debt ratio actually drops in 2050 by more than 10 percentage points.⁹ Similarly, Pisani-Ferry and Mahfouz (2023) explore debt dynamics for decarbonization scenarios of revenue and spending policies in France. They also assume that the government will provide significant support to households and firms to cover part of the decarbonization investments, especially in the building sector. In their central scenario, the debt-to-GDP ratio increases by 25 percent points of GDP in 2040. About half of this increase is associated with public expenditures; another significant portion of the rise in debt is caused by the loss in tax revenue due to a slowdown in economic growth in their model.¹⁰

While political-economy considerations are critical for the design and implementation of carbon mitigation policies, political red lines might be relaxed through carefully designed policy packages (including compensation measures and clear communication) and through international coordination. For instance, governments can make carbon pricing more palatable by emphasizing the fairness of the polluter-pay-principle and by designing comprehensive policy packages that compensate the most vulnerable households and most affected firms and workers for their impact. International cooperation on carbon pricing could mitigate effects on competitiveness, relocation of industries, and carbon leakage. The rest of this paper explores in more detail how a global side agreement (as a complement to the Paris Agreement) and forms of compensation could shape the fiscal equation of countries.

IV. A GLOBAL AGREEMENT ON DECARBONIZATION

A global side agreement on carbon mitigation should be based on several principles. First, it should include at least the largest countries in the world. Agreement among the G20, for example, would cover over 80 percent of global emissions. But a smaller group comprising China, the EU, India, the US would still cover more than 60 percent of global GHG emissions. When complemented with the African Union (AU) (which recently joined the G20 as a member), it would also cover a great deal of the world's diversity

⁹ A revenue neutral approach, perhaps with subsidies financed through higher property taxes, can be more efficient as it avoids a net fiscal cost on the government and prevents favoring the housing sector over other sectors.

¹⁰ A meta-analysis in the October 2023 Fiscal Monitor shows a wide variety of findings on the GDP effects of climate mitigation policies. While simulation-based studies often report (small) negative effects, recent empirical studies suggest mostly positive or insignificant effects.

and establish a clear link with the global development agenda.¹¹ Second, a side agreement should contain concrete policy actions (like minimum pricing requirements) to deliver the needed global emissions reductions. Since carbon pricing might not be feasible in all countries, the agreement should allow for flexibility in instrument choice to accommodate alternative (but equivalent) approaches (Black et al. 2022a). Third, the side agreement should address international equity issues. Those are reflected in the Paris Agreement through the principle of common but differentiated responsibilities, which would entail different requirements for emission reduction. For developing countries these requirements will have to be compatible with energy access and in line with the Sustainable Development Goals. Moreover, side payments in the form of financial and technological transfers and/or revenue sharing of a globally agreed minimum carbon price will likely be part of an international agreement.

Based on these principles, we consider a concrete scenario that closes the global mitigation gap to be consistent with 2°C warming (see Black, Parry, and Zhunussova 2023c). It assumes that high- and upper-middle income countries¹² agree to reduce their emissions in 2030 by 49 percent and 26 percent, respectively, below 2019 levels. Low-income and lower-middle income countries would still be able to grow their emissions by 6 percent relative to 2019, but this is less than implied by current mitigation pledges and the baseline. The scenario brings about near convergence in per capita CO₂ emissions among high and higher-middle-income countries while emissions remain lower in lower-middle and low-income countries. While the scenario assumes global implementation, the paper focuses on results for members of the G20, including the African Union. Within the G20, India, Indonesia and the African Union are captured under the group of lower-middle income countries. We also present separate results for 3 large members of the African Union: Egypt, Kenya, and Nigeria.

The IMF-World Bank Climate Policy Assessment Tool (CPAT) is used to simulate the policy scenario (see Black et al. (2023a) for full documentation of the model).¹³ CPAT is a flexible model providing projections for all countries of their future fuel use and CO₂ emissions by major energy sectors (power, transportation, industry, buildings, agriculture). It can simulate the impact of a diverse range of pricing and non-pricing mitigation policies on CO₂ emissions and on fiscal, distributional, and economic variables. CPAT is not a structural model built on micro-economic foundations or specific technologies but rather is a streamlined (or ‘reduced form’) model that captures the main aspects of climate policies in significant detail. Baseline emissions projections depend on GDP growth, income elasticities for energy products, autonomous improvements in energy efficiency, and reductions in the cost of renewables. In the baseline scenario, fuel taxes/subsidies and carbon prices are held fixed at current levels in real terms. The responsiveness of emissions to pricing and other policies depends on induced changes in energy prices and fuel price responsiveness in different sectors. Parameters are set at the mid-range of the empirical literature on

¹¹ Global collective decision making might involve a mix of representatives from nation-states and supra-national organizations, such as the African Union, the EU, ASEAN, or the Organization of American States, see e.g., Tim Murithi in “Orders of Oppression” in Foreign Affairs May/June 2023. The G-20 constitutes a precedent as the EU is a member (alongside several EU member states).

¹² For the purposes of this paper, we define country groups in accordance with the [World Bank Classification](#) which includes four income groups, the thresholds for the lowest group (LIC) being \$1,135 per capita GNI in 2022 and for the second lowest (Lower-middle income countries, LMIC) being \$4,465. It is different from the [IMF’s World Economic Outlook](#) classification (which distinguishes between advanced economies and emerging and developing economies).

¹³ See also CPAT documentation here: [CPAT Documentation \(cpmodel.github.io\)](#).

income and fuel price elasticities and emissions-price responsiveness from more detailed, country-specific energy models. The focus is on the impact in 2030.

Emission reductions could efficiently be achieved through carbon pricing (taxation or an emission trading scheme) (see next section). However, alternative price or non-pricing measures could also be used, such as feebates, standards, regulations, or subsidies, possibly applied to specific sectors like industry, power, buildings, transport, forestry, and extractives. The assumption here is that policies are efficient in the sense of minimizing the cost of emission reduction. This would hold under a single carbon price but not necessarily under non-pricing approaches. The next section discusses assumptions about an agreed carbon price floor to achieve the climate objective (Parry, Black, and Roaf 2021).

Using CPAT, we first estimate the impact of these efficient policies on marginal abatement costs in 2030—reflecting the shadow carbon price of induced policies at the efficiency frontier (left panel of Figure 3). The shadow carbon prices are estimated at around \$145, \$90, and \$35 per ton in high, higher-middle, and low and lower-middle-income countries, respectively. The global weighted average is around \$90 per ton.

Shadow prices are calculated in CPAT by the CO₂ price that induces the target emissions reductions. These prices are truncated at \$150 per ton because uncertainty about marginal abatement costs rapidly increases beyond this point. For instance, it will depend on non-linear adoption of uncertain ‘breakthrough’ technologies, like carbon capture and storage for large stationary emitters and direct air capture. Truncation at \$150 implies some underestimation of total abatement costs in countries where the actual marginal abatement costs will exceed \$150. This applies to several high-income countries in Figure 3. Raising the threshold to \$200 or \$250 per ton and linearly extrapolating the marginal abatement cost curve would increase the total costs for EU-27 by, respectively, 12 and 17 percent.

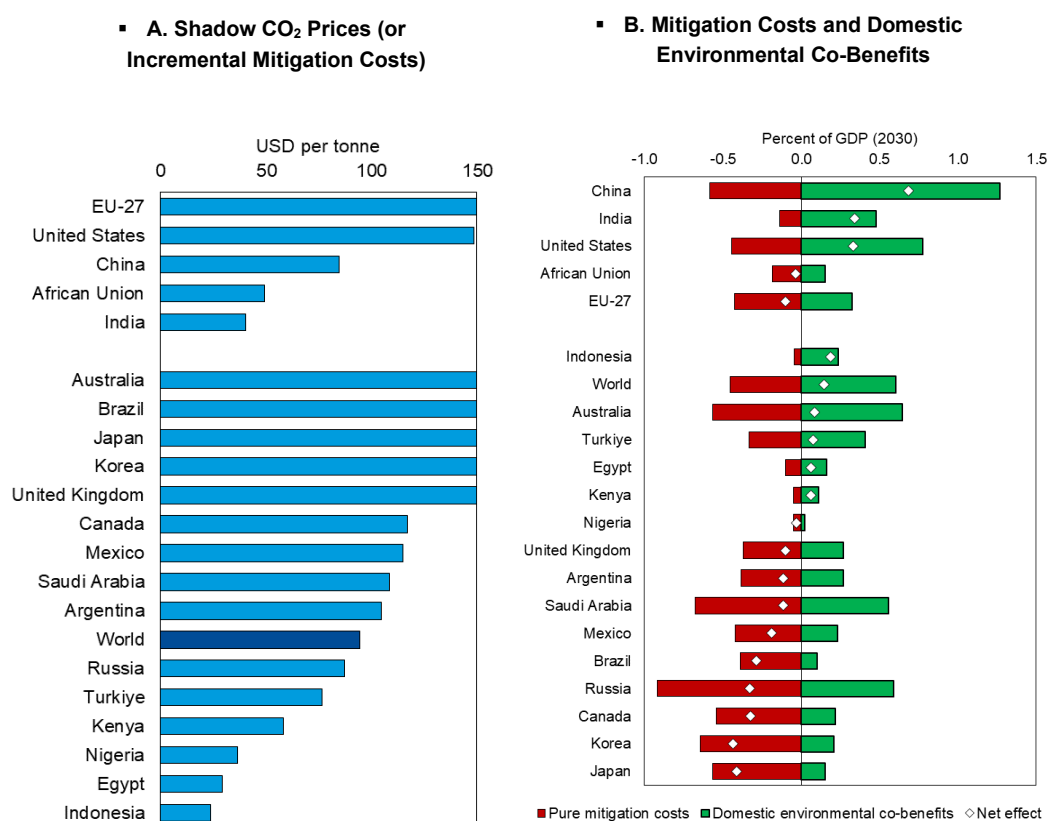
The right panel in Figure 3 shows the total abatement costs, or welfare costs, of these policies. These are approximately the annualized costs of using cleaner but costlier technologies (in the model, based on the areas under the marginal abatement cost schedules).¹⁴ Total abatement costs are estimated at around 0.5 percent of GDP at the global level. Costs are somewhat higher in high-income countries, where cheaper mitigation options have been largely exhausted. The total costs are well below 0.5 percent of GDP in low- and lower-middle income countries, reflecting the availability of cheaper mitigation options and less stringent emissions reductions. The lowest costs are observed in India, Indonesia, Egypt, Nigeria, and Kenya. Costs might become higher, of course, if less efficient mitigation policies would be used that do not strike a cost-effective balance across mitigation responses. Moreover, for high-income countries they may also be higher to the extent that marginal abatement costs exceed the \$150 ceiling.

The right panel of Figure 3 also shows the effects on so-called domestic environmental co-benefits. In particular, the burning of fossil fuels not only leads to climate change, but also causes local air pollution. This causes premature deaths, more instances of disease and undesirable health impacts (e.g., strokes

¹⁴ Equivalently, welfare costs reflect losses in consumer and producer surplus in fossil fuel markets net of government revenue gains, and accounting for pre-existing fuel taxes or subsidies. Welfare costs are a standard metric used in cost/benefit assessments of environmental and broader fiscal/regulatory policies. The cost estimates however do not account for mitigation in forestry and agriculture nor revenue losses to energy exporters. And they can mask significant distributional impacts, including the shift of resources from emissions intensive to cleaner production. GDP measures differ from welfare costs as they include changes in net exports and investment.

and lost pregnancies) and reduces labor productivity. Fossil fuel use in transportation, moreover, contributes to congestion, accidents, and other transportation externalities. Climate mitigation policies can reduce these externalities, yielding localized welfare gains. CPAT quantifies their value, based extensive data on such impacts and information about localized effects. The right panel in Figure 3 shows that domestic environmental co-benefits will offset a portion of the higher mitigation costs of climate policy in several countries. For some countries, such as China, India, and Indonesia they even exceed the abatement costs. Hence, these countries are better off from a decarbonization policy, even before accounting for climate benefits. This does not hold for the average country in the African Union, where abatement costs are slightly higher than the local environmental co-benefits.¹⁵

Figure 3. Incremental and Total Mitigation Costs from Equitable Ambition Scenario for 2°C



Source: Black, Parry, and Zhunussova (2023c).

Notes: Prices and annual costs are measured relative to a baseline with no new, or tightening of existing, mitigation policies. Shadow CO₂ prices are truncated at \$150 per ton. Actual marginal abatement costs will likely be higher, although uncertainty in this range of the cost curve is large. It implies that average abatement costs will be higher in countries where the shadow price is binding (which holds for most Advanced Economies).

¹⁵ The co-benefits vary substantially across countries with, for example, variations in local air emission rates, population exposure to local pollution, and people's willingness to pay to lower health risks. In Saudi Arabia a disproportionately large share of the CO₂ emissions reductions comes from reduced use of road fuels where other externalities (like congestion and accidents) per unit of fuel use are large.

V. PUBLIC REVENUE

Carbon pricing is the centerpiece of an efficient mitigation policy. A carbon price internalizes the external cost of carbon emissions, promoting the full range of behavioral responses for reducing the emissions intensity, including through investment in clean technology, fuel substitution, and energy saving. Aside from boosting the adoption of existing low-carbon technologies, carbon prices also encourage the development of new technologies through innovation. A uniform price of carbon implies that the marginal abatement cost of each alternative behavioral response to emissions reduction is equal. This ensures that the policy is efficient as total abatement costs are minimized. This is not necessarily the case with regulations, subsidies, or sectoral policies. Carbon pricing can be implemented through a tax on the carbon content of fossil fuels, or through an emission trading scheme (ETS) where permits can be auctioned. It can raise significant amounts of public revenue, which we refer to as the ‘fiscal dividend’.¹⁶ Revenue from carbon pricing has attracted ample attention, especially since it could be used productively to contain the overall costs of decarbonization for the economy. For instance, carbon tax revenues could provide a new source of funding for high-return public investments or for cutting other distortionary taxes that harm incentives to work and invest.¹⁷

We explore the fiscal dividend in the 2°C decarbonization scenario outlined in the previous section using CPAT. Effects are presented relative to a baseline that includes only current policies. We consider two pricing scenarios. The first scenario is implemented by using only carbon pricing (“Full carbon price”) to achieve the agreed emission reduction. Thus, carbon prices are equivalent to the shadow prices presented in Figure 3. The second scenario uses two policy instruments. First, we set carbon prices at a globally agreed minimum level (“Capped carbon price”) of \$75, \$50, and \$25 per ton of CO₂ for, respectively, high-, middle-, and lower-middle and low-income countries in 2030.¹⁸ While significantly reducing emissions, these carbon prices are insufficient to meet the required targets aligned with a 2°C temperature increase. We therefore assume pricing is supplemented by additional (revenue neutral) policies that mitigate the remainder of the emissions, such as feebates, standards, and regulations.

Figure 4 shows estimates of the fiscal dividend in 2030 for both pricing scenarios relative to the baseline. The estimates capture the revenue effects of not only the carbon tax itself, but also the base erosion of existing fuel taxes on road and other fossil fuels (or subsidy in some cases). Decarbonizing road transport by shifting to electric vehicles will progressively erode the existing tax bases of motor fuel excises.

The left panel of Figure 4 shows that gross revenues from full carbon taxes in 2030 range between 0.5-3 percent of GDP for most countries. For several high-income countries, such as countries in Europe and

¹⁶ Governments can also employ alternative pricing schemes that forego some or all revenue, such as an ETS with free allocation of allowances or feebates. The focus of this section is on the fiscal dividend from carbon pricing schemes.

¹⁷ Governments should first consider phasing out fossil fuel subsidies if they still have them. Black et al. (2023b) find that explicit subsidies (undercharging of the supply costs of fossil fuels) globally cost \$1.3 trillion in 2022, up from \$0.5 trillion in 2020. Their phase out would imply a significant cost saving for governments across the globe, while eliminating the perverse incentive to consume more fossil fuels. Black et al. (2023b) also quantify implicit subsidies, measured as the undercharging of external costs of fossil fuel use (climate externalities and other, domestic environmental externalities). If the total subsidies would be removed, the primary balance for governments would improve by \$4.4 trillion globally.

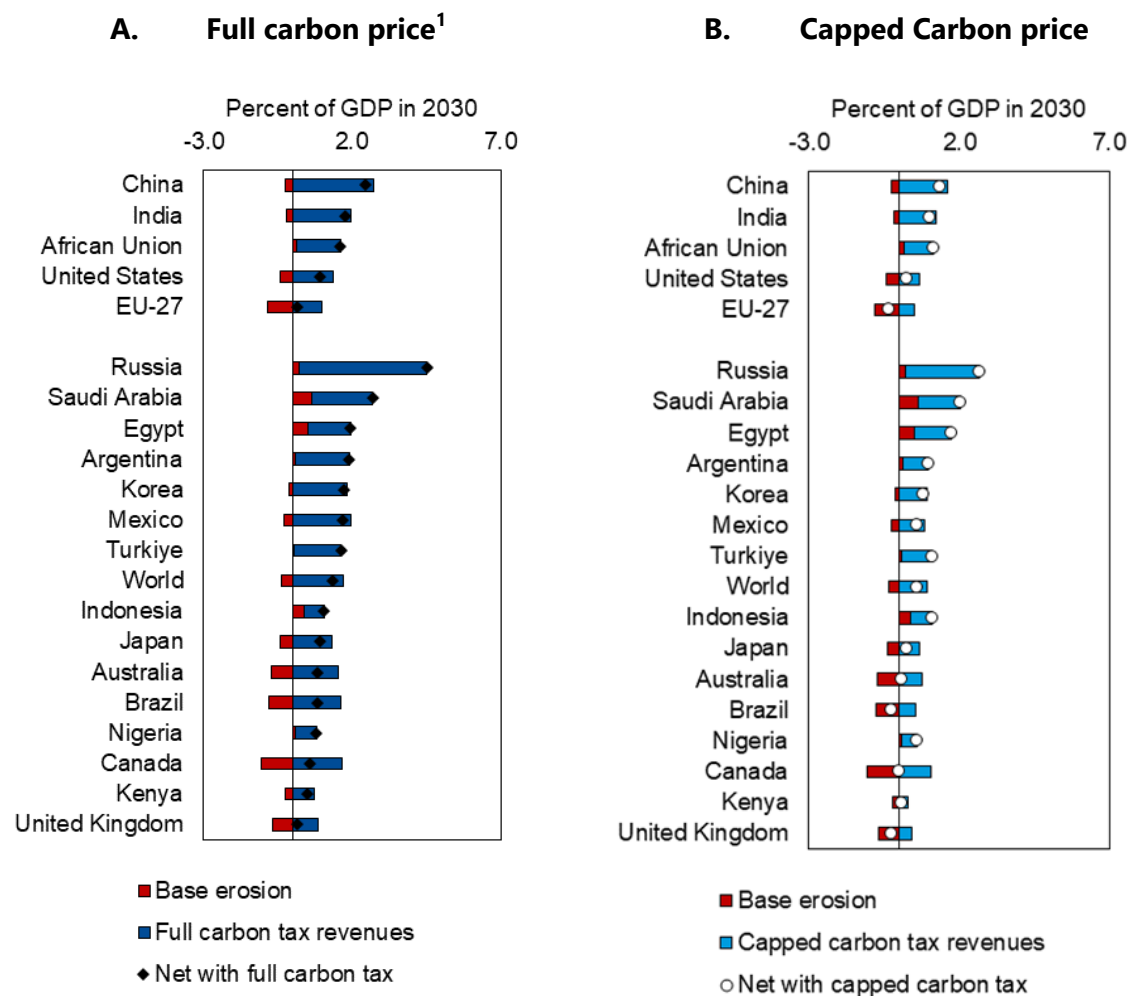
¹⁸ This follows the proposal for a global minimum carbon price floor in Parry, Black, and Roaf (2021). The “capped” carbon pricing achieves 67 percent of the reduction needed to stay within 2°C.

the US, prices are close to \$150 per ton; revenue typically ranges between 1 and 1½ percent of GDP. Prices in middle and low-income countries are often lower, ranging from less than \$50 per ton in Nigeria, Egypt, and Indonesia up to \$150 per ton in Brazil and Korea. Revenue varies from less than 1 percent of GDP in Kenya, Nigeria, and Indonesia to 2 percent or more in China, India, Saudi Arabia, and Mexico.

The right panel in Figure 4 shows gross carbon revenue under the capped carbon price, which is lower than the full carbon price. Hence, gross revenue from this carbon tax is also lower, especially since the emission reduction is the same due to the assumed non-pricing policies. Gross carbon revenue in Europe is generally around ½ percent of GDP, while in the US and Japan it is slightly higher at ¾ percent of GDP. In most middle and low-income countries, revenue is somewhat higher, at around 1 percent of GDP. This may seem surprising as the carbon price in high-income countries is higher than in middle- and low-income countries. The reason for this difference in revenue effects is that the emissions intensity of GDP in high-income countries tends to be relatively lower, in part as a result of current policies. Revenue potential under the capped carbon price is found to be the largest in China and Russia, where it exceeds 1.5 percent of GDP. In countries of the African Union, revenue is around 1 percent of GDP on average.

The red bars in Figure 4 show revenue loss from base erosion of existing fuel taxes. In most countries, revenue from existing fuel taxes will decline in 2030. Currently, road fuel taxes typically raise between 0.5 – 1.5 percent of GDP across G20 countries. In countries with more aggressive targets for replacing internal combustion engine vehicles (ICEVs) with electric vehicles (EVs), these fuel tax bases will erode by significantly by 2030. In Figure 4, we see that the erosion causes a decline in fuel tax revenue between 0.2 and 1.1 percent of GDP. However, in seven cases (Indonesia, Egypt, Russia, Saudi Arabia, African Union, Argentina, Nigeria), fuels are currently subsidized. For these countries, the switch to EVs will imply an additional fiscal gain.

Figure 4. Fiscal Impacts of Policies in 2030 to Close Ambition Gaps and Stay on Track with Net Zero Targets



Source: IMF staff calculations using CPAT.

Note: "Full carbon price" assumes that the carbon price is equivalent to the shadow price presented in Figure 3. This scenario is equivalent to Black, Parry, and Zhunussova (2023c). "Capped carbon price" assumes a carbon price of \$25 for low and lower-middle income countries, \$50 for upper-middle income countries and \$75 for high-income countries; remaining emission reduction to achieve the target is assumed to be achieved by non-pricing instruments.

Under an alternative approach where emission targets would be met entirely with non-pricing instruments, there would be no fiscal dividend from carbon pricing. Non-pricing instruments include (tradeable) emission rate standards and their fiscal analogue, feebates. If these instruments are applied and harmonized across the power, industry, transport and building sectors they would promote many of the same behavioral responses as carbon pricing. But the revenue gains shown in the blue bars in Figure 4 would disappear. At the same time, the revenue loss from base erosion of existing fuel taxes (the red bars) would be approximately the same. Hence, this decarbonization scenario would imply a significant revenue loss for almost all countries. To offset this revenue loss, countries may seek alternative revenue

from transportation. For instance, countries could adopt a charge per vehicle km traveled as a way to secure a more robust revenue source from transportation. At the same time, such policy would internalize transport-related externalities like traffic congestion.

VI. PUBLIC SPENDING

Some forms of government spending will be necessary as part of an efficient decarbonization scenario. This includes spending on green public infrastructure, public support for innovation, and possibly support for deployment of existing low-carbon technology where market imperfections are important.

Investments in infrastructure networks may benefit multiple users and would be undersupplied when left to the market. Examples are pipelines for hydrogen and carbon capture and storage, high voltage transmission lines to link up different sites for renewable plants, and EV charging stations.¹⁹ With a share of public investment in total of approximately 15 percent in high-income countries, this would imply a fiscal cost of around 0.2 percent of GDP in the coming decades, based on IEA (2022) estimates. In middle- and low-income countries, however, the current share of public investment is around 60 percent as the power sector in these countries is often dominated by greater state-ownership and electricity prices are regulated. Hence, a larger share of investments in renewable plants may be reflected in government accounts. Yet, the public share in the future is highly uncertain and depends on the pace of privatization and developments in the cost of low-carbon technologies.²⁰ As a larger share of investment gets borne privately and countries liberalize energy markets over time, the public costs would fall. Moreover, even if a significant share of the power sector remains in public hands, it would still be more efficient to reflect investment costs in higher prices for users (e.g., higher electricity tariffs) rather than subsidize the investments from public budgets. This would help promote the efficient allocation of capital and other resources across sectors rather than favoring the power sector over other sectors.

Private sector efforts to **innovate and deploy low-carbon technologies** may be inefficiently low because individual firms do not capture possible benefits to other firms, such as knowledge spillovers from developing or pioneering low-carbon technologies. For instance, other firms can imitate or use embodied knowledge to further their own technology or benefit from ‘learning-by-doing’ at early adopters by hiring former workers or imitating production techniques. Scale economies may also deter manufacturing of immature technologies.

Government intervention to support innovation can come in different forms. First, governments should invest in **basic research** for unproven or costly technologies that may ultimately be essential for the low-carbon transition, such as direct air capture. Given the global public good nature of basic research, international coordination may help scale up such investments to account for positive cross-border

¹⁹ Estimates for public investments in a decarbonization scenario will likely feature a hump-shaped pattern over time (IMF 2021): the increase is concentrated over the next 20 years and will likely decline thereafter. For example, renewable energy has a high upfront investment cost in the form of solar panels and windmills, but low operating costs as no fossil fuels are needed. Similarly, new network infrastructure needs to be constructed like high voltage transmission lines to renewable sites or electric vehicle charging stations. Note that promoting energy efficiency can help limit infrastructure needs.

²⁰ The October 2023 Global Financial Stability Report of the IMF assumes that the private share in low- and middle-income countries will rise in the coming decades to between 85 and 90 percent (IMF 2023b).

knowledge spillovers. Fiscal outlays for basic research into clean energy technologies have been estimated at around \$30 billion per year globally (IEA 2020). A gradual ramping up of such research funding might be needed, along with an expansion of the supply of scientists and engineers in this area (Newell 2015).

Second, governments, at least in high-income countries, play an important role in **promoting private R&D** considering the positive externalities this creates for others (IMF 2016). Governments commonly do this through intellectual property right protection, research subsidies, R&D tax credits, and occasionally technology prizes. In principle, with a robust and efficient price of carbon emissions, additional incentives for clean technology should be similar to those for general R&D. However, more generous treatment for green technologies can be warranted if the appropriability problem is more severe than for other R&D. This may be plausible for technologies that are currently far from the market (e.g., green hydrogen-based energy). The general fiscal outlays for R&D, exogenous subsidies and R&D tax credits in high-income countries are modest, however, at around 0.2 percent of GDP²¹ (somewhat higher in Canada, France, UK and the US, and somewhat lower in Germany and Scandinavia). Only a small fraction of these general R&D incentives is geared toward low carbon technologies.

Finally, governments can provide assistance for **technology deployment**, such as for EVs, heat pumps, energy storage, green hydrogen, carbon capture and storage, or wind and solar energy (although part of these might be captured under public investment, e.g., if power companies are public). In the power sector, wind and solar have already become competitive with fossil fuel technology and offer a promising path toward net zero emissions. Nevertheless, public intervention may still be warranted to accelerate their dissemination to be in line with emissions targets. Subsidies and tax incentives for these technologies are heavily concentrated in a few key countries, such as China, Italy, Japan, Korea, Germany and the US, but they are generally below 0.2 percent of GDP.

While these spending policies have recently gained interest, they are also contentious due to various inefficiencies. First, they come along with a fiscal cost that could lead to fiscally unsustainable outcomes, as was illustrated in Section III. If higher debt is avoided by raising other taxes, this will exacerbate pre-existing tax distortions in labor and capital markets and increase the efficiency costs of climate policies. To minimize these inefficiencies, it would therefore be best to apply subsidies and tax credits incrementally and avoid windfall gains for investments that would have been made also without the policy. Moreover, subsidies and tax incentives should be temporary and removed over time as technologies mature and become competitive. Second, subsidies and tax credits encourage substitution toward clean energy but do not incentivize energy savings, like carbon taxes do. This renders them less efficient.²² Third, subsidies and tax credits can distort international markets, especially if provisions discriminate against foreign manufacturers through domestic content requirements. However, even without such discrimination they can induce green companies to relocate to countries with larger tax incentives and subsidies, at the expense of countries without such benefits. A technological race has a winner-takes-all logic that can exacerbate geopolitical tensions and protectionist sentiments. Finally, like traditional industrial policies, green subsidies and tax credits are prone to government failures associated with rent-

²¹ See [OECD R&D tax incentives database](#).

²² For instance, Bistline, Mehrotra, and Wolfram (2023) find that the marginal abatement cost of the US Inflation Reduction Act ranges between \$45-\$61, while a \$15 uniform carbon price could have achieved the same overall emission reduction.

seeking and corruption. Thus, they could lead to wasted resources, competitiveness losses or state capture if not well-implemented.

VII FISCAL COMPENSATION

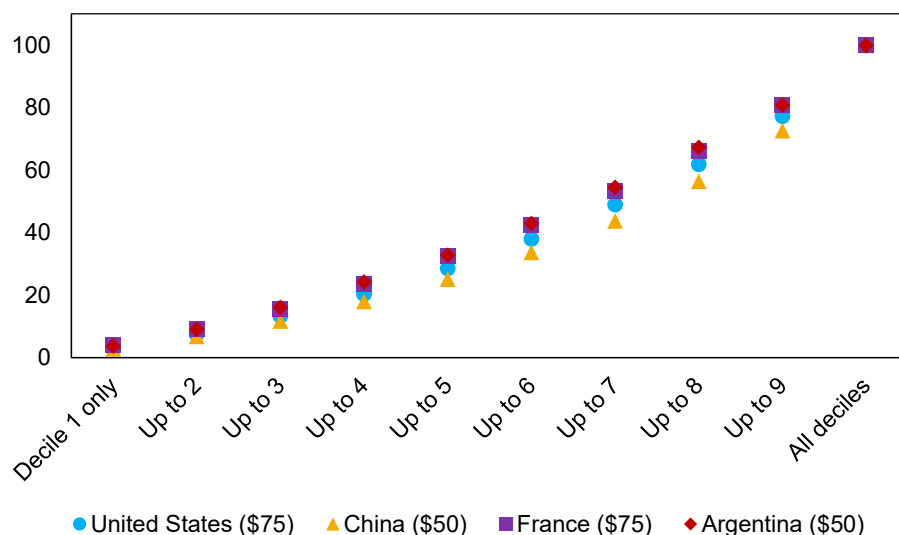
Compensation schemes will be necessary to achieve a politically viable policy mix. This will be especially so in the case of carbon pricing schemes, where decarbonization will impose an additional burden on households and firms prior to revenue recycling. This section discusses the fiscal impact of these compensation schemes.

Carbon pricing will have negative implications for households' purchasing power by directly raising the cost of energy. For instance, the IMF (2019) finds that a \$75 per ton carbon tax would, on average across countries, triple the use price of coal, raise natural gas prices by 70 percent, that of electricity by 45 percent, and gasoline by 13 percent.²³ Moreover, these higher energy prices can indirectly raise the prices of non-energy goods that use fossil fuels as an input in their production. In high-income countries, the effects on real household incomes tend to be slightly regressive or distribution neutral (IMF 2019). In middle- and low-income countries, however, they are usually found to be neutral or slightly progressive. In either case, household impacts can be mitigated by recycling the revenues through strengthening social protection and/or by lowering taxes elsewhere. If revenues are used for targeted compensation of the bottom income deciles, the overall reform can be made progressive—and efficient if compensation takes the form of cuts in distortionary taxes (for example, payroll taxes) rather than transfers (Klenert et al. 2018).

We use CPAT to analyze how much of the carbon price revenue is required to offset the impact on households in different income deciles in four countries: Argentina, China (both for a \$50 carbon price), France and the US (both for a \$75 carbon price). The estimates on household real incomes include both policy-induced increases in energy prices and higher prices for other consumer goods (using country specific input-output tables and matching to consumer budget shares for different products from household surveys). Burdens are generally regressive (i.e., larger relative to consumption for lower income than higher income households) but only moderately so. Figure 5 suggests that targeted transfer to compensate the three bottom income deciles will require around 15 percent of the carbon price revenue in each of the four countries. Compensating the bottom half of households requires approximately 30 percent.

²³ Energy prices can fluctuate significantly, and did notably so in recent years. The price increases reported here are based on projections for 2030, using extrapolations from data for the late 2010s (see IMF 2019).

**Figure 5. Costs to Fully Compensate Household Deciles for the Impact of a Carbon Price
(% of revenue raised)**



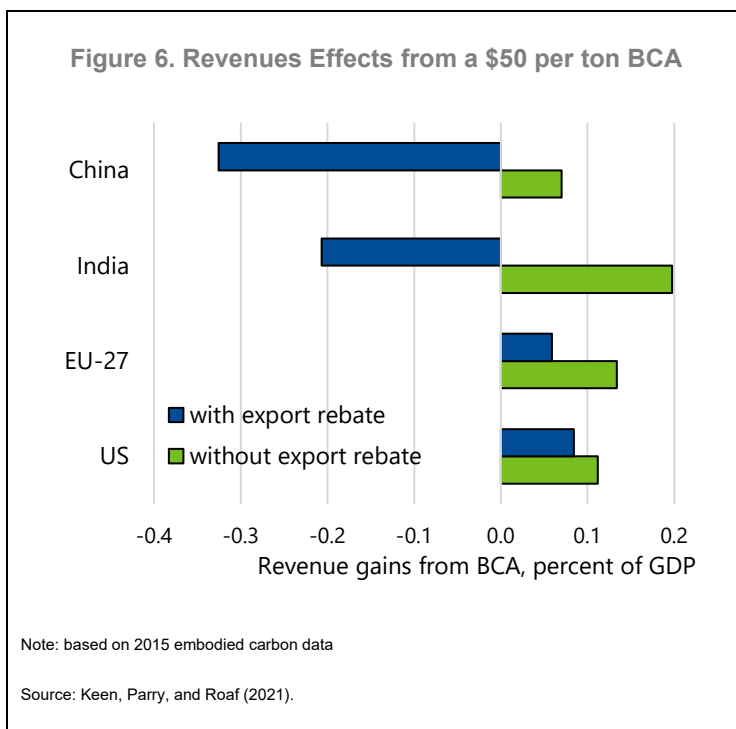
Source: CPAT

Note: The transfer is calculated to fully offset the higher tax burden on households up to the respective income decile and expressed as a share of overall revenue from the carbon price. Carbon price for France and the US is assumed to be \$75 and for Argentina and China \$50.

Compensation might also be needed for certain firms. A global agreement on climate mitigation would help sustain a level playing field across countries and thus reduce the need for countries to take unilateral measures to maintain competitiveness of industries in response to domestic climate action. Moreover, a global deal would mitigate concerns about carbon leakage (counteracting increases in emissions abroad due to international migration of production), since all firms in energy-intensive trade-exposed industries would be affected by similar mitigation measures. However, in the absence of a global side agreement, there will be calls for offsetting measures if a country unilaterally imposes a carbon price. Indeed, unlike subsidies or regulations, carbon pricing imposes a charge on remaining (non-abated) emissions, which could negatively impact production costs for energy-intensive trade-exposed industries. This could be addressed by combining carbon pricing with partial assistance to firms (providing free allowance allocations, broader tax reliefs, subsidies), or by using other approaches (tradable performance standards or feebates). This, however, would forego the fiscal dividend and, if pursued through subsidies, even involve an additional fiscal cost.

When countries unilaterally adopt carbon pricing, they can also use border carbon adjustments (BCAs) for energy-intensive trade-exposed industries to mitigate competitiveness effects. BCAs are charges on embodied carbon in imports, potentially complemented by rebates for embodied carbon in domestic exports, in order to offset the impact of domestic carbon pricing. They have the potential to level the playing field in domestic markets (for imports) and foreign markets (when rebates are provided on exports). Moreover, they mitigate carbon leakage effects and may provide incentives for other countries to impose similar carbon prices to preempt the impact of the BCA. Intuitively, taxing carbon emissions on a destination basis—as opposed to an origin basis—

does not induce production distortions. However, BCAs need to be designed and administered carefully to be effective (Parry, Black, and Roaf 2021). The potential revenues from BCAs on imports are estimated at around 0.1-0.2 percent of GDP for a \$50 per ton charge—for China, India, the EU, and the US (Figure 6). With export rebates, net revenues are found to be negative for China and India. From a broader perspective, however, taxing carbon on a destination basis leaves more scope for countries to exercise their domestic ambitions without hurting competitiveness. For instance, BCAs may enable higher carbon prices and thus higher revenue. The revenue losses from export rebates might be small compared to the overall revenue gain from comprehensive pricing of domestic carbon emissions.



VIII. INTERNATIONAL REVENUE SHARING

In 2009, advanced economies pledged to mobilize US\$100 billion a year from 2020 for mitigation and adaptation in developing economies, both from private and public sources. Estimates suggests that flows have not yet reached this level, however. Developing economies are likely to face much higher investment needs for adaptation to climate change, i.e., investments to make countries resilient against the impact of climate change.

Financial transfers from advanced to developing countries will form an integral part of an international side agreement on decarbonization. The transition to a low-carbon economy might be costly for developing countries with rapidly growing energy demand and fiscal and financial constraints limiting options for funding green public investments or domestic compensation measures.

There are various approaches to financial transfers and sharing the burden of decarbonization between countries. For instance, a normative approach proposed by Rajan (2021) starts from the assignment of equal property rights to access the global common. If climate externalities are efficiently internalized by making polluters pay for the social cost of carbon through a globally agreed carbon price, the global

revenue from such a charge could be distributed across the global population on a per-capita basis to reflect the equal distribution of property rights. If carbon prices are initially collected by individual countries based on their own emissions, poor countries should then be compensated by rich countries through financial transfers. Indeed, fiscal transfers would depend on the difference between a country's own emissions per capita (on an origin basis) and the world average: countries with emissions per capita below the global average (usually poor countries) would be net receivers, while countries with higher emissions per capita (usually rich countries) will be net payors. This would link the climate agenda directly to the broader development agenda and the sustainable development goals.²⁴

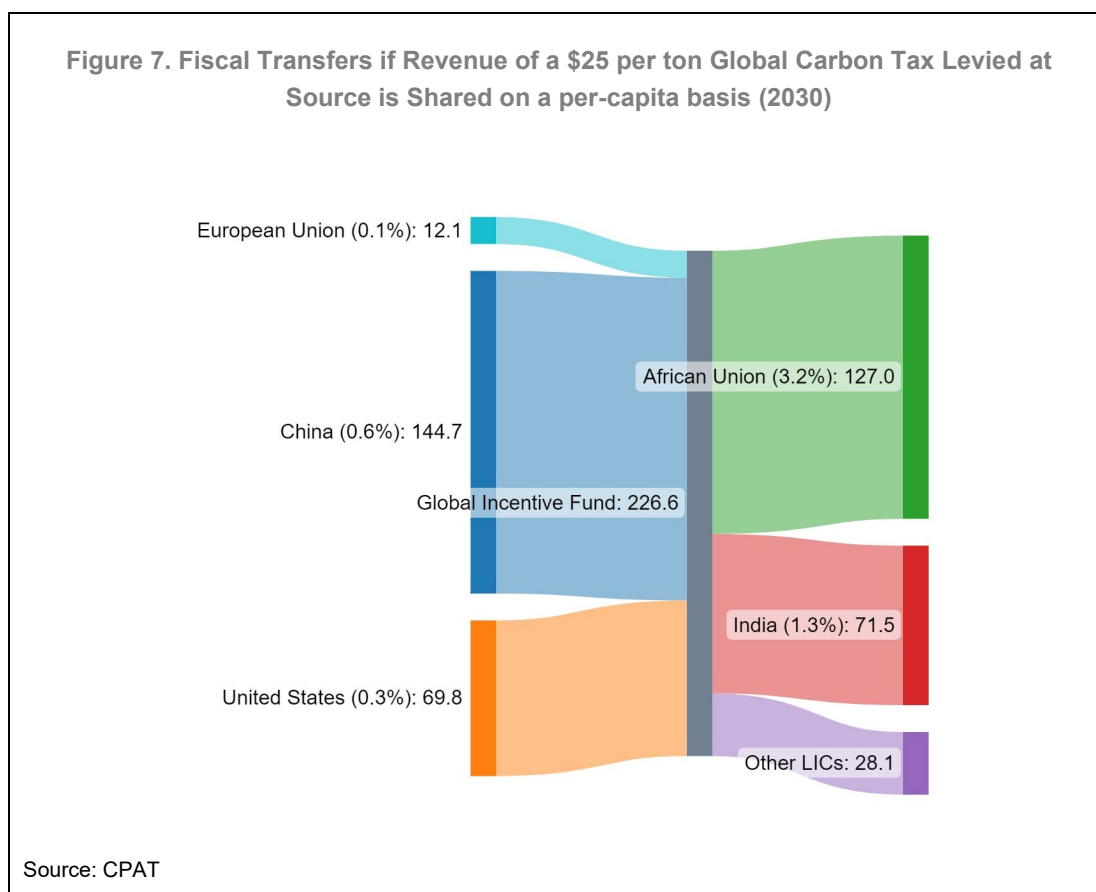


Figure 7 uses CPAT to simulate the magnitude of these transfers under a \$25 per ton carbon tax in 2030 for five countries or country groups (EU, African Union, China, US, India). We see that approximately \$230 billion would be transferred to low- and lower-middle income countries, as they feature carbon emissions per capita below the world average. Out of the total, \$127 billion would go to the African Union (3.2 percent of their GDP) and \$71 billion to India (1.3 percent of GDP). Net payors would be countries

²⁴ Revenue sharing is perhaps most salient for carbon pricing in international shipping and aviation, as it is not straightforward to allocate taxing rights to either origin, destination, residence, or transit countries. For a \$75 per ton carbon price, global revenue from international transportation could be around \$100 billion in 2030 (Black, Parry, and Zhunussova 2023c).

with carbon-per-capita above the global average, which includes China (0.6 percent GDP), the US (0.3 percent GDP) and the EU (0.1 percent GDP).

This scenario is merely illustrative. Many other possibilities exist. For example, the scheme could also operate on a destination basis (in line with an overall system of carbon taxation complemented with carbon-border adjustment).

IX. PUTTING PIECES TOGETHER

The net fiscal impact of global climate action for individual countries will depend on how policies affect the different pieces of the fiscal equation. Fiscal revenues increase if mitigation is done through carbon pricing – which is more likely to emerge forcefully under global action than under uncoordinated unilateral action. However, revenue from pre-existing taxes on motor fuels are currently significant in many high-income countries and will decline. On the spending side, there are various needs to support decarbonization, such as public infrastructure, public research, stimulus for private R&D, and possible deployment subsidies. Also transfers will be necessary to compensate domestic citizens and firms. Financial transfers from high to low and lower-middle-income countries will further impact the fiscal equation of individual countries.

Figure 8 illustrates the magnitude of the net effects on the fiscal equation in the decarbonization scenarios analyzed in this paper for a selection of G20 countries in 2030. Given the uncertainty and arbitrary nature of some assumptions about public spending, we consider two public spending scenarios: a high-spending and a low-spending scenario (see details in Figure 8).²⁵ Moreover, we consider both the scenario with full carbon pricing to achieve global climate targets and the scenario with a capped carbon price, supplemented by non-pricing policies.

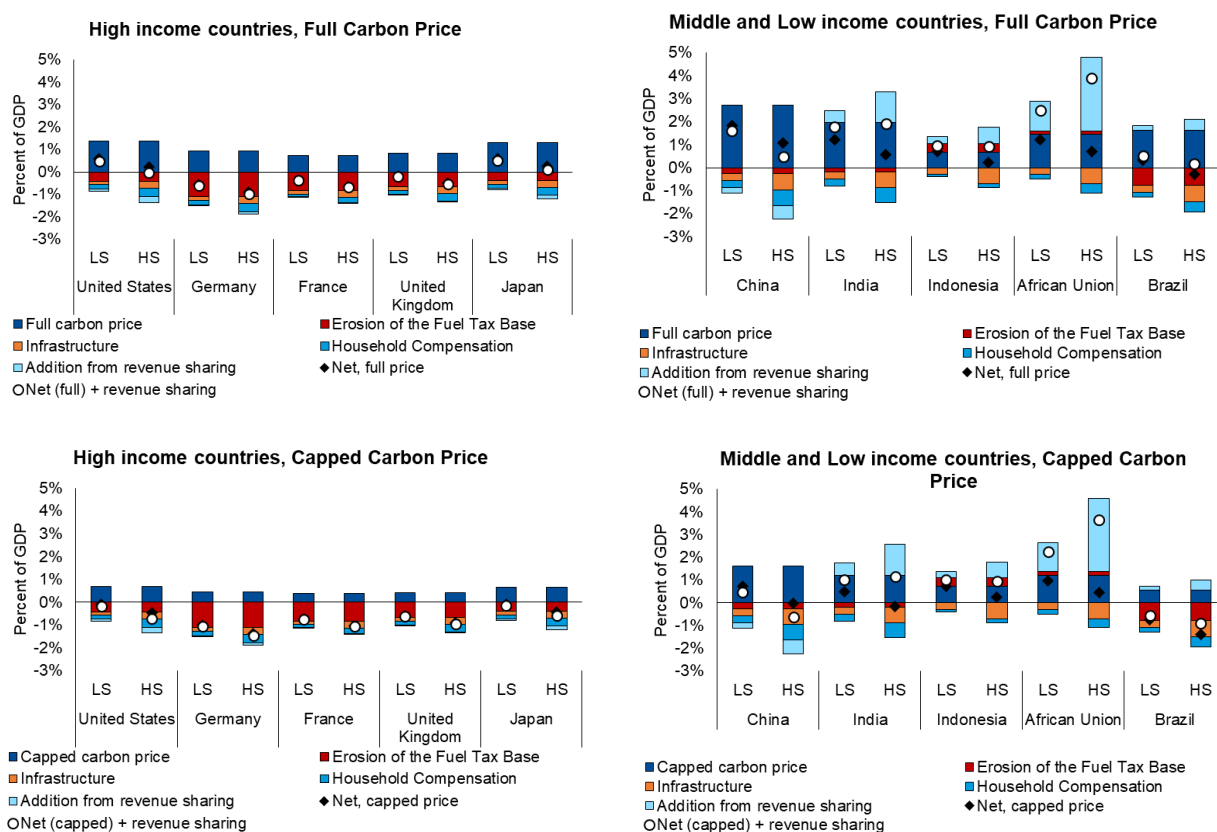
In European countries, we see that revenue (from carbon pricing and fuel tax) changes on balance by between -0.2 and 0.1 percent of GDP in 2030 in the full pricing scenario. Even where the net revenue effect is positive, this is insufficient to cover the higher expenditures on green public infrastructure, subsidies, transfers, and compensation to low-and lower-middle-income countries – irrespective of the low or high spending scenario. A cap on the carbon price further deteriorates the fiscal equation for these countries. Hence, European countries may see their debt rise or they need to raise other taxes, such as road-user charges, to offset negative effects on the primary balance.

In the US and Japan, revenue gains are closer to 1 percent of GDP in the full pricing scenario. This is higher than in Europe due to higher carbon-intensity of GDP. In the low-spending scenario, this revenue gain renders the impact on the fiscal balance positive. Hence, these countries could reduce pre-existing taxes as part of a revenue-neutral reform, or they could reduce their public debt burden. In the high-spending scenario, however, the impact on the fiscal balance is close to neutral. If the carbon price is capped, the fiscal balance is close to neutral under the low-spending scenario but deteriorates under the high-spending scenario.

²⁵ The numbers presented here are illustrative and should be interpreted with caution. They are based on specific assumptions about policies and reflect scenarios that could happen, rather than normative scenarios of optimal policies. Public spending needs in each group are assumed to be the same as a percentage of GDP across countries but will differ in practice.

Revenue gains are generally positive in middle and low-income countries and largest in China and India, where carbon revenue (under full carbon pricing and without base erosion) is more than 2 percent of GDP. In the African Union and Brazil, carbon revenue rises by around 1.5 percent of GDP under full pricing, while it is close to 0.7 percent of GDP in Indonesia. The revenue side is further augmented in the African Union, India, and Indonesia by revenue sharing of a global carbon price equal to \$10 or \$25 in the two spending scenarios. This revenue sharing also improves the fiscal balance in Brazil. However, it reduces it for China due to its relatively high carbon intensity.

Figure 8. Impact of Decarbonization Scenario on the Fiscal Balance in Selected Countries in 2030 (Low Spending (LS) and High Spending (HS) Scenario) and Full and Capped Carbon Pricing



Note: Carbon tax revenue and erosion of the fuel tax base are based on Figure 4 under both full and capped carbon pricing. Low spending scenario (LS) assumes additional spending on infrastructure and subsidies by 0.15% GDP in advanced economies and 0.3% in emerging and developing economies; transfers compensate the bottom 3 deciles (0.15% of carbon revenue), and revenue sharing of \$10 carbon price. High spending scenario (HS) assumes additional spending on infrastructure and subsidies by 0.3% in advanced economies and 0.7% in emerging market and developing economies; transfers compensate bottom 5 deciles (0.3% of carbon revenue), and revenue sharing of \$25 carbon price.

In India, Indonesia and the African Union, the revenue gains in the full carbon price scenario exceed the additional expenditures – both under low- and high-spending and even in the absence of revenue sharing. Hence, these countries could use the additional fiscal space to support the wider development agenda, e.g., for adaptation investment or other sustainable development goals. In China, the fiscal

balance improves in the absence of revenue sharing and in the low-spending scenario. This improvement largely disappears, however, if the carbon price is capped at \$50 and spending is high. While the same applies to India, revenue sharing would turn the net fiscal impact positive also under high spending and capped pricing. Finally, for Brazil the effects on the fiscal balance are small and can be somewhat negative if the carbon price is capped, revenue sharing is absent and with high spending. Revenue sharing, however, will turn it into a marginally positive fiscal impact if carbon pricing is not capped.

X. CONCLUSION

This paper starts by discussing the three-way trade-off that countries face when unilaterally managing their climate transition, namely between meeting their climate policy objectives, maintaining debt sustainability and respecting political constraints. High climate spending, as recently pursued in several advanced economies through subsidies and public investment programs, can support country's climate goals while avoiding political resistance. However, such a scenario is likely fiscally unsustainable and can be inefficient. In contrast, relying primarily on carbon pricing will produce favorable fiscal outcomes and achieve carbon mitigation in an efficient manner. Yet, carbon pricing can be unpopular, and this scenario could thus cross political red lines. Scaling back climate ambition could relax both fiscal and political constraints but will violate climate commitments—and thus come at a large economic cost.

Countries can relax this policy trilemma through a carefully designed package that includes a mix of different policies and which is based on an international side agreement. While carbon pricing forms the centerpiece of this policy package, it is complemented by other regulatory and spending policies as well as compensation schemes that support political acceptability. International cooperation helps maintain a level playing field and addresses concerns about competitiveness, relocation of industries, and carbon leakage. The international agreement should include at least the largest countries in the world (China, EU, India, US); contain concrete policy actions (like minimum pricing requirements or equivalent approaches) and should address international equity issues to link climate mitigation to the wider development agenda.

The paper then analyzes a concrete policy scenario of global carbon mitigation, which includes both coordinated carbon pricing, other mitigation policies, compensation schemes, and a revenue sharing mechanism that breaches the financial divide between rich and poor countries. The scenario is designed to achieve a path toward 2°C above pre-industrial temperatures, as agreed in Paris in 2015. The analysis focuses on the implications for the fiscal equation of a diverse group of countries.

We find that fiscal revenues depend on the balance of rising carbon revenue and the gradual erosion of existing fuel tax bases. While this balance is positive for most countries, some advanced European countries will face a negligible or even negative net revenue balance in 2030. Public spending rises during the transition to build green public infrastructure, promote innovation, support clean technology deployment, and compensate households and firms. Estimates of these spending needs vary considerably, however, and depend on speculative assumptions about the (changing) role of government in several countries. On balance, while global decarbonization will have negative fiscal implications for most European countries, effects can be positive for the US and Japan if public spending is contained. For middle and low-income countries, net fiscal impacts are generally positive and sometimes significant—mostly due to more buoyant revenue effects that exceed spending. For low-income countries, these effects are reinforced if a portion of the global revenue from carbon pricing is shared across

countries on a per-capita basis. As this global mitigation strategy improves fiscal balances in developing countries, it supports the global development agenda.

ANNEX. COMPARING THIS PAPER WITH THE FISCAL MONITOR

This paper uses the CPAT framework to assess the fiscal implications of a global side agreement on carbon mitigation. It presents different scenarios for carbon pricing policies and considers different assumptions for public spending. The IMF's Fiscal Monitor, released in October 2023, also shows the fiscal implications of different climate mitigation scenarios.²⁶ However, those scenarios are not part of a global side agreement and assumptions for the policy scenarios are different. It also uses a different economic model to focus on debt dynamics over a longer time horizon. Key parameters in the model used in the Fiscal Monitor are calibrated for an average advanced or emerging market economy, while CPAT uses country-specific information. Table A.1 shows the main differences between this paper and the Fiscal Monitor – both in terms of the assumptions about the policy scenarios and the methodology to assess its impact.

Table A.1: Comparing analysis in this paper and the Fiscal Monitor

	This Paper: Global Side agreement	Fiscal Monitor: Unilateral Policy
Analysis/Methodology		
Model	Climate Policy Assessment Tool	New Keynesian Dynamic General Equilibrium Model
Years	2030	Years between 2023 – 2050
Countries	Calibrated for 200 countries, focus on G20, including African Union	Calibrated for a representative large, advanced economy and a large emerging market economy
Baseline	Current policy	Current policy
Fiscal variables	Fiscal balance	Primary balance, interest, and public debt
Feedback mechanisms on fiscal balances	None	Endogenous GDP and interest rates
Policy assumptions		
Carbon prices	Capped carbon price: Increase by \$75, \$50 and \$25 per ton of carbon in, respectively, high-income countries, higher-middle-income countries and low-and lower-middle-income countries – non-pricing policies achieve remaining reduction to achieve agreed reduction.	Advanced Economies: increase gradually from currently \$40 to \$130 in 2030 and \$235 in 2050 (\$280 in alternative scenario) Emerging Market Economies: increase gradually from currently \$5 to \$45 in 2030 and \$150 in 2050 (\$165 in alternative scenario)

²⁶ Details of the model and the policy scenario can be found in on-line appendix 1.2 of the Fiscal Monitor, see [here](#).

	Full carbon price: increase in the carbon price to achieve agreed global emission reduction.	
Fuel taxes	Base gradually erodes in 2030	Base assumed to be replaced by miles-per-vehicle charge to offset impact of eroding fuel tax base
Household compensation	Low-spending scenario: 15 percent of carbon tax revenue (lowest 3 deciles) High-spending scenario: 30 percent of carbon tax revenue (lowest 5 deciles)	30 percent of carbon tax revenue
Public infrastructure + subsidies	High-income countries: Low-spending scenario 0.15 percent of GDP; high-spending scenario 0.3 percent of GDP Low and middle-income countries: Low-spending scenario 0.25 percent of GDP; high-spending scenario 0.7 percent of GDP	Advanced economy: 0.5 percent of GDP Emerging market economy: Low-spending scenario: 0.25 percent of GDP; High-spending scenario 0.75 percent of GDP
International revenue sharing	Low-spending scenario: per capita revenue sharing of \$10 per ton carbon tax in 2030 High-spending scenario: per capita revenue-sharing of \$25 per ton carbon tax in 2030	None

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